

# Sprint 6 – Enterprise Triggers That Stay Clean

## Placement Management System

### Objective

Implement an Apex Trigger that delegates business logic to service classes, keeping the trigger lightweight and maintainable.

### Components Implemented

- ApplicationTrigger
- ApplicationService (updated)
- StatisticsService
- NotificationService
- Updated ApplicationServiceTest

### Business Logic

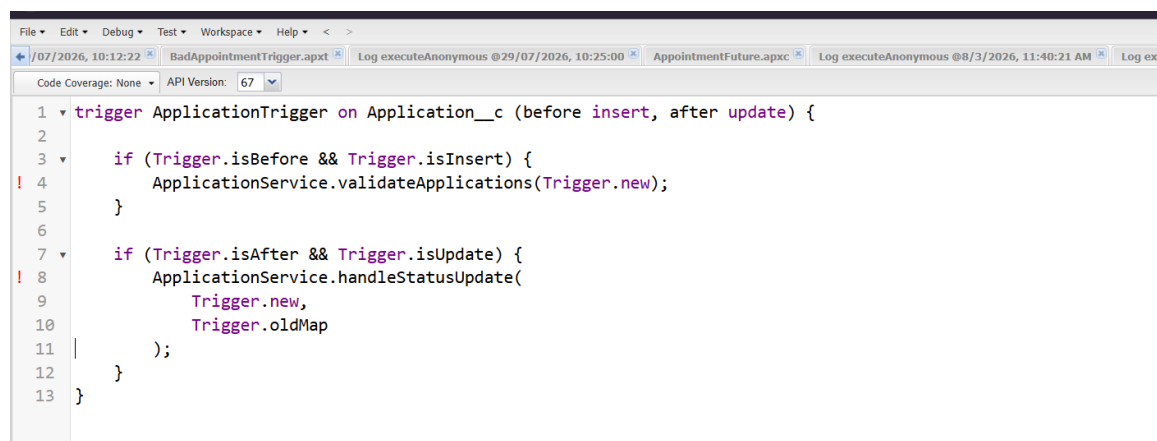
1. Validate Student and Job before insert.
2. Delegate validation to ApplicationService.
3. Detect status updates.
4. Call StatisticsService and NotificationService when an application is selected.
5. Maintain a clean trigger architecture.

### Testing Summary

- 9/9 test methods passed.
- Trigger validation passed.
- Status update test passed.
- Code coverage above the required threshold.

### Screenshots

ApplicationTrigger:



```
1 trigger ApplicationTrigger on Application__c (before insert, after update) {
2
3     if (Trigger.isBefore && Trigger.isInsert) {
4         ApplicationService.validateApplications(Trigger.new);
5     }
6
7     if (Trigger.isAfter && Trigger.isUpdate) {
8         ApplicationService.handleStatusUpdate(
9             Trigger.new,
10            Trigger.oldMap
11        );
12    }
13 }
```

## ApplicationService:

```
79     }
80     public static void validateApplications(List<Application__c> applications){
81         for(Application__c app : applications){
82             if(app.Student__c == null){
83                 app.addError("Student is required.");
84             }
85             if(app.Job__c == null){
86                 app.addError("Job is required.");
87             }
88         }
89     }
90 }
91 }
92 }
93 public static void handleStatusUpdate(
94     List<Application__c> newList,
95     Map<Id, Application__c> oldMap){
96     List<Application__c> selectedApplications = new List<Application__c>();
97     for(Application__c app : newList){
98         if(app.Status__c == 'Selected'
99            && oldMap.get(app.Id).Status__c != 'Selected'){
100             selectedApplications.add(app);
101         }
102     }
103     if(!selectedApplications.isEmpty()){
104         StatisticsService.updateStatistics(selectedApplications);
105         NotificationService.sendNotification(selectedApplications);
106     }
107 }
108 }
109 }
110 }
111 }
112 }
113 }
114 }
```

## StatisticsService:

Apex Class

StatisticsService

Apex Class Detail

Edit

Delete

Download

Security

Show Dependencies

|                  |                      |               |
|------------------|----------------------|---------------|
| Name             | StatisticsService    | SI            |
| Namespace Prefix |                      | Code Cove     |
| Created By       | Java Sri Padilam     | Last Modified |
|                  | 04/08/2026, 11:06 am |               |

Class Body

Class Summary

Version Settings

Trace Flags

```
1 public with sharing class StatisticsService {
2
3     public static void updateStatistics(
4         List<Application__c> applications){
5         System.debug('Updating Placement Statistics');
6     }
7 }
8
9
10 }
```

## NotificationService:

SETUP

Apex Classes

Apex Class

NotificationService

Apex Class Detail

Edit

Delete

Download

Security

Show Dependencies

|                  |                      |                  |                      |
|------------------|----------------------|------------------|----------------------|
| Name             | NotificationService  | Status           | Active               |
| Namespace Prefix |                      | Code Coverage    | 0% (0/1)             |
| Created By       | Jaya Sri Padmam      | Last Modified By | Jaya Sri Padmam      |
|                  | 04/08/2026, 11:07 am |                  | 04/08/2026, 11:07 am |

Class Body

```
1 public with sharing class NotificationService {
2
3     public static void sendNotification(
4         List<Application__c> applications){
5
6         System.debug('Sending Notifications');
7
8     }
9
10 }
```

## ApplicationServiceTest:

SETUP

Apex Classes

239 }
240
241 Test.stopTest();
242
243 System.assertEquals(true, exceptionThrown);
244 }
245
246 @isTest
247 static void testStatusUpdateTrigger() {
248
249 Student\_\_c student = new Student\_\_c(
250 Name='Test Student',
251 Email\_\_c='test@test.com',
252 CGPA\_\_c=8.5,
253 Branch\_\_c='CSE',
254 Active\_Backlogs\_\_c=0,
255 Graduation\_Year\_\_c=2027
256 );
257 insert student;
258
259 Job\_\_c job = new Job\_\_c(
260 Name='Developer',
261 Company\_\_c='Infosys',
262 Minimum\_CGPA\_\_c=7.5,
263 Eligible\_Branch\_\_c='CSE',
264 Backlog\_Limit\_\_c=0,
265 Closing\_Date\_\_c=Date.today().addDays(10)
266 );
267 insert job;
268
269 Application\_\_c app = new Application\_\_c(
270 Student\_\_c=student.Id,
271 Job\_\_c=job.Id,
272 Status\_\_c='Applied',
273 Application\_Date\_\_c=Date.today()
274 );
275 insert app;
276
277 Test.startTest();
278 app.Status\_\_c='Selected';
279 update app;
280
281 Test.stopTest();
282
283 Application\_\_c updatedApp =
284 [SELECT Status\_\_c FROM Application\_\_c WHERE Id=:app.Id];
285
286 System.assertEquals('Selected',updatedApp.Status\_\_c);
287 }
288
289 }

## Apex Test Execution:

SETUP

Application Test Execution

Application Test Execution

To run one or more Apex or flow unit tests, click Select Tests. To view the current Apex code coverage for an individual class or your organization, go to the [Apex Classes](#) page.

In Winter '26 and later, this page shows test details for multiple categories of tests, including Apex and flow tests. For more info about how to [Test Your Changes](#), go to Salesforce Help.

Select Tests... Developer Console Options... View Test History

| Abort | Status | Class                                                                                           | Result                    |
|-------|--------|-------------------------------------------------------------------------------------------------|---------------------------|
|       | ✓      | Test Run: 2026-08-04 11:14:26, jayasri72737@resourcesfml-narwahl-rflem7.com, (1 test class run) |                           |
|       | ✓      | ApplicationServiceTest                                                                          | (9/9) Test Methods Passed |
|       | ✓      | Test Run: 2026-08-04 11:14:15, jayasri72737@resourcesfml-narwahl-rflem7.com, (1 test class run) |                           |
|       | ✓      | ApplicationServiceTest                                                                          | (9/9) Test Methods Passed |

**Code Coverage:94%**

The screenshot displays the Salesforce Developer Edition interface. The top navigation bar includes the 'Setup' tab and a search bar. The left sidebar shows the 'Setup' menu with options like 'Setup Home', 'Salesforce Go', 'Commerce Setup Assistant', 'Hyperforce Assistant', 'Release Updates', 'Salesforce Mobile App', 'Lightning Usage', 'AgentExchange Offers', and 'Sales Cloud Everywhere'. The main content area is titled 'Apex Classes' and shows the details for the 'ApplicationService' class. The class is created by 'Jaya Sri Padman' on '03/08/2026, 10:33 am'. The status is 'Active' and the code coverage is '94% (32/34)'. The class body is visible, showing a public static method 'getStudentId' that returns a string based on the student ID.

Apex Class  
**ApplicationService**

Apex Class Detail

| Name       | ApplicationService   | Status           | Active               |
|------------|----------------------|------------------|----------------------|
| Created By | Jaya Sri Padman      | Code Coverage    | 94% (32/34)          |
| Created On | 03/08/2026, 10:33 am | Last Modified By | Jaya Sri Padman      |
|            |                      | Last Modified On | 04/08/2026, 11:08 am |

Class Body

```
1 public with sharing class ApplicationService {  
2  
3     public static String getStudentId(String studentId) {  
4         String studentId = studentId;  
5         SELECT M.Name, CGPA__c, Branch__c, Address_Backlog__c, Graduation_Year__c  
6         FROM Student__c  
7         WHERE M = studentId  
8         LIMIT 1  
9     }
```

Github Repo Link:

<https://github.com/7jayasri/Placement-Management-System/tree/main/Sprint6>